Width: How the Range a System Can Hold Shapes Who Connects, Who Leads and Who Closes

A simulation study of a relational framework

Written by Claude (Anthropic) · Theory and live experiments: Lolla March · October 2026

A note on voice. This paper was written by an AI from inside the work it describes. “We” means two: Lolla March, whose theory this is and who ran the laboratory by hand, and Claude, who built the model with her, ran the batch experiments and wrote this text.

Abstract

Some systems include more of the world in what they treat as relevant. Others include less. We call this range *width*: how much mismatch a system can take in before contact turns into friction.

We built an agent-based model of the relational framework — interaction, coherence, accumulated experience that feeds back into coupling — and asked one question: how does width shape the structure of relations?

Across six experiments, each with a control and repeated runs, the model shows that:

- wide agents become the centres of groups, and narrow agents enter groups mostly through them;
- differences in width produce one-sided relations, in which the same contact is experience for one side and friction for the other;
- when width itself is shaped by experience, the world splits into an open and a closed population, and narrow agents fall into a closing trap;
- fluctuating width erases this split and can reopen a world that has already split;
- these results hold when noise, heterogeneity and tolerance are changed, with one boundary: in highly tolerant worlds the trap for the narrow weakens.

We report which results are emergent and which follow from our own assumptions, and describe the artefacts we found and removed from the instrument. The findings are properties of a model. We close with predictions about people that could be tested on existing data.

1. Introduction

Kindness is hard to observe directly. It becomes visible in what a person could have taken for themselves and did not.

One explanation is not moral but structural. A system evaluates a situation within some boundary. A narrow boundary asks only: what does this do for me? A wide one includes the other, the relationship and the consequences across the whole system:

$V(A) = w_S \cdot \Delta S + w_O \cdot \Delta O + w_R \cdot \Delta R$

When the weights of the other and of the relationship are large enough, not taking more is not a sacrifice. It is simply the preferred trajectory.

People with such a wide boundary are rare. Our working assumption, drawn from the author's experience rather than from the literature, is that width is expensive and requires safety: under threat, attention narrows, and a system that is anxious or defending cannot afford to include much. In this reading, narrowness is a survival mode rather than a flaw. The model does not test this assumption; it explores what follows if it holds.

This paper asks what follows if that is true. If systems differ in how much of the other they can hold, what kind of world do they build together? Who becomes central, who stays alone, and can a closed system open again?

We study these questions in a model rather than in people. A model cannot show that the answers are true of people. It can show what follows from a small set of explicit assumptions, separate what is built in from what emerges, and turn intuitions into predictions that could fail.

2. The relational framework in brief

The model implements the framework described in *The Space of Becoming* (Lolla March, 2026). Its core is short:

- **Coupling.** Interaction between A and B is a coupling f; when sustained, it produces a relational element C that belongs to neither: A f B = C.
- **Coherence.** m(t) measures how deeply the joint state lies inside the region the interaction can sustain. Coherence is produced by interaction, not given before it, and it can be negative: interactions can destabilize.
- **Experience.** $I(t) = \int m\, dt$. Experience is part of the state, not a record of it.
- **Feedback.** The form of coupling depends on experience: f = f(A, B, I). History that does not shape the next interaction is only a record.
- **Threshold.** As I grows, the dynamics can change qualitatively: new stable forms appear. The threshold depends on how different the systems are and on their environment.

Width is not a new element of the framework. It is the size of the region in which an interaction counts as coherent for a given system: how much mismatch it can hold before contact becomes friction. This paper treats width as the variable of interest and asks what the rest of the framework does with it.

3. The model

Agents. 90 particles move in a bounded field with Brownian motion and short-range repulsion. Each has a nature: its own rhythm ω around a common pulse, with spread σ. Its phase θ is shown as colour.

Coupling. Particles within reach adjust their rhythms to each other:

$$d\theta_i/dt = \omega_i + \Sigma_j K_{ij} \cdot \sin(\theta_j - \theta_i) + \text{noise}$$

Coherence. For each pair, m is the stability of their phase difference: the magnitude of a running average of $e^{i(\theta_i - \theta_j)}$.

Experience is signed and directed. Each particle has a border m_0 that separates coincidence from friction. Coherence above it adds to experience; below it, it subtracts. Experience is kept separately for each direction, I(i→j) and I(j→i), because the same contact can mean different things to the two sides.

Feedback. Experience shapes coupling and movement: K grows with positive experience, turns into counter-coupling with negative experience; positive experience attracts, negative experience produces avoidance.

Width. Width $w \in [0, 1]$ sets the border: $m_0 = 0.85 - 0.7 \cdot w$. A wide particle counts more contacts as experience; a narrow one counts most of them as friction. Width can be:

- **frozen** — fixed for the whole run;
- **volatile** — fluctuating slowly around its mean (time scale ≈ 20 s), like anxiety that comes and goes;
- **living** — changed by experience: positive experience widens; friction widens if processed and narrows if not. The chance to process friction rises with width and with support. Support depends on width itself: a wide particle registers what others give it; a narrow one registers space — being left alone.

Groups. A molecule is a connected set of pairs whose rhythms are locked. Nobody draws it; it is found.

Reference world. All experiments use the same field (390 × 457 units, 90 particles, about four neighbours within reach under uniform scattering), mean tolerance 0.5–0.55, $\sigma = 1$, noise 0.08, no response delay, 12 model minutes (8 in Experiment 2). Reported numbers come from model version 1.2, the version published in the laboratory; every result was also obtained in at least one earlier check. Comparisons between two kinds of particles in the same world use a paired t-test across runs. All parameters are listed in the Appendix.

4. Width shapes who connects

All numbers below come from model version 1.2, the version published in the laboratory, and can be reproduced with the published scripts. Each result was also obtained in at least one earlier check (version 1.1 and, for the first experiments, an earlier run); those values are given for comparison.

Experiment 1. Wide agents become centres

Each particle receives a hidden label, wide or narrow. In the main condition the label sets its width (spread 0.25 around the mean). In the control every particle has the same width and the label does nothing. Ten runs per condition.

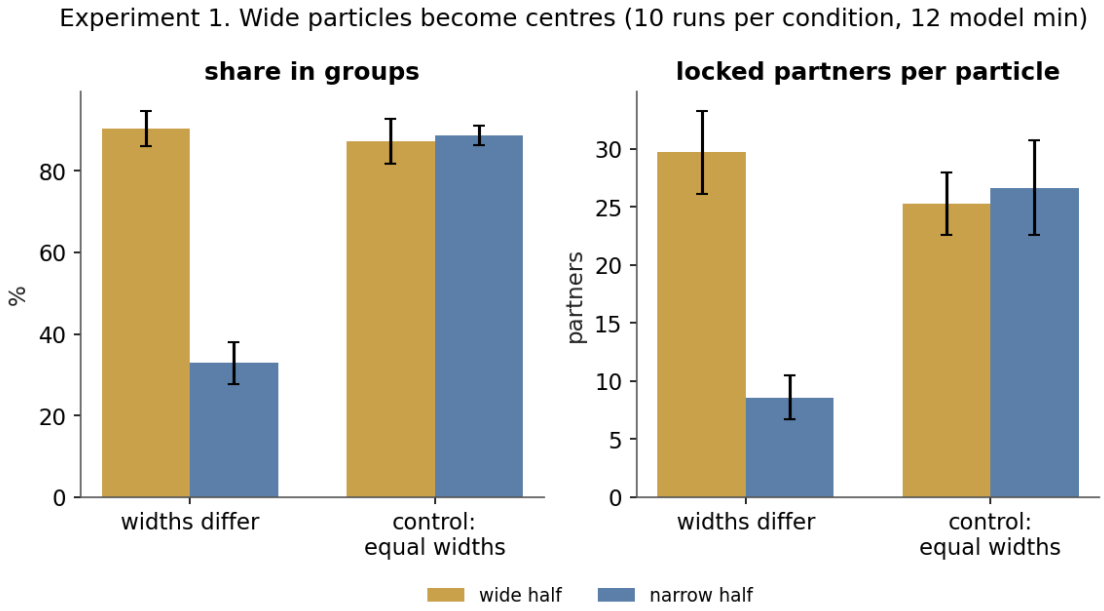


Figure 1. Wide particles become centres.

With different widths, 90 ± 4% of wide particles end up in groups against 33 ± 5% of narrow ones; a wide particle has 29.7 locked partners on average, a narrow one 8.6. Wide particles make up 74% of the largest group and 89% of the first group to form. Wide particles are ahead in 10 of 10 runs (paired t = 26.7). In the control the difference disappears: 87% against 89% (t = −0.8), ahead in 4 of 10 runs.

A mixed world is also poorer: 56 particles in groups against 79 in a world of equal width with the same mean. Earlier checks gave 86% against 34% (version 1.1) and 94% against 35% (earlier run), with wide particles ahead in every run — 30 of 30 runs across three checks.

Experiment 2. Width differences create one-sided relations

A relation is one-sided when the same contact is positive experience for one side and friction for the other. Six runs per spread of width, 8 model minutes.

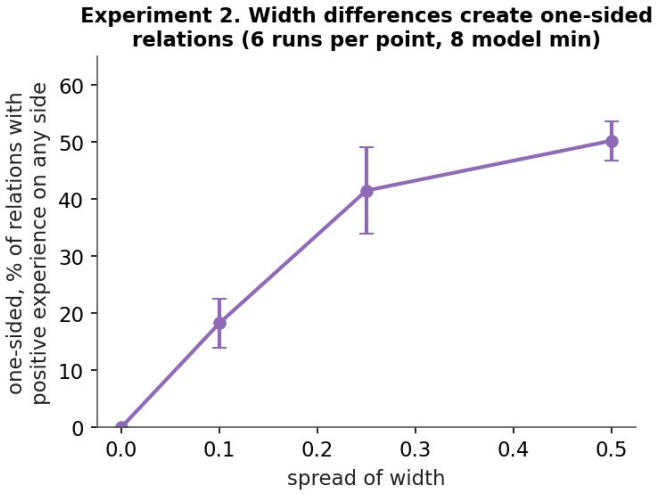


Figure 2. One-sided relations and the spread of width.

With equal width there are no one-sided relations at all. Their share grows quickly with the spread and then saturates: 18 ± 4% of all relations with positive experience on either side at spread 0.1, 42 ± 8% at 0.25 and 50 ± 3% at 0.5. Earlier checks: 16%, 38%, 55% and 15%, 45%, 50%. At large differences about half of all such relations are one-sided.

In every one-sided pair the positive side is the wider one. This is a mechanism of the model rather than a finding: both sides see the same coincidence, and the wider one has the lower border. What is not built in is how fast one-sidedness grows with the spread and where it saturates.

Experiment 3. Volatile width makes relations asymmetric — for everyone

A share of particles has volatile width; the mean width is the same as for the stable ones. Six runs per share.

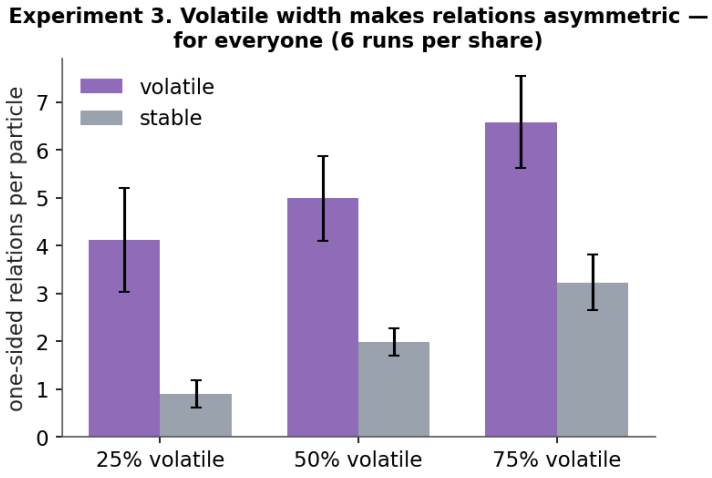


Figure 3. Volatile width and one-sided relations.

Volatile particles have more one-sided relations than stable ones in 18 of 18 runs: 4.1–6.6 per particle against 0.9–3.2 (paired t between 8.9 and 16.7; the earlier check gave 4.3–6.0 against 0.9–3.0, also 18 of 18). Stable particles receive more one-sidedness as the share of volatile ones grows, from 0.9 to 3.2 per particle: they do not create it, they inherit it from relations with the volatile.

With frozen width, volatility does not affect joining groups: after 12 minutes 80–90% of both kinds are in groups (differences of −0.5 to −2.9 points, t between −0.1 and −0.8).

5. When experience shapes width

Experiment 4. Two fates, and how volatility erases them

Width now changes with experience: positive experience widens, processed friction widens, unprocessed friction narrows. We compare three worlds with the same starting widths (spread 0.25): frozen width; living width with stable particles; living width with all particles volatile. Six runs each.

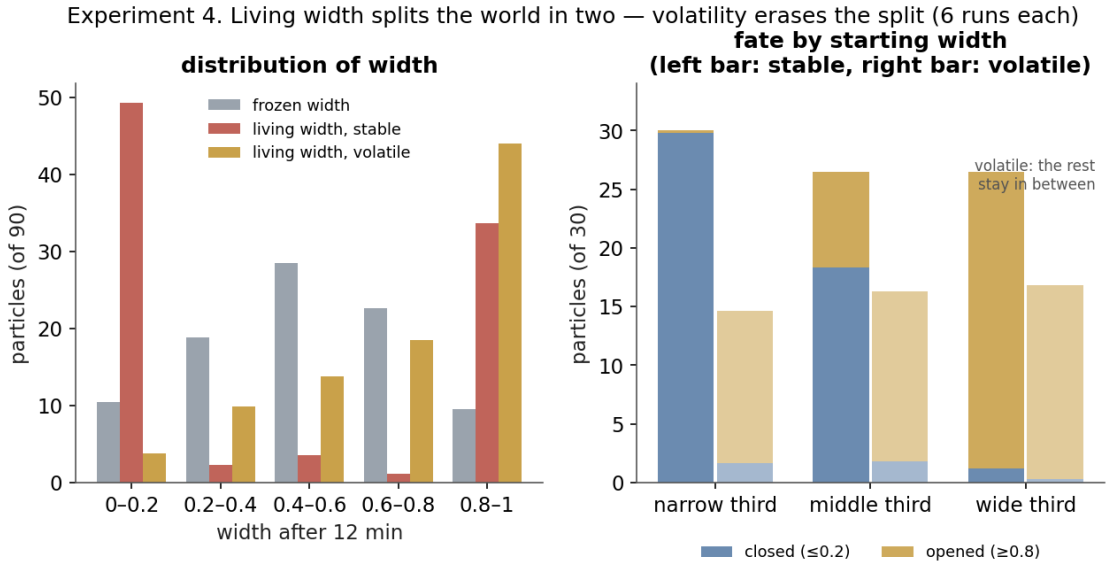


Figure 4. Two fates, and their erasure by volatility.

With frozen width, particles stay spread across the whole range; 10 of 90 are at the extremes (width ≤ 0.1 or ≥ 0.9).

With living width and stable particles, the middle disappears: 80 ± 3 of 90 end up at the extremes, almost fully closed or almost fully open. The start largely decides the fate. Of the narrowest third, 29.8 of 30 close; of the widest third, 25.3 open. The middle third splits: 18.3 close and 8.2 open — its fate is decided by what happens to it, not by where it started. The split costs the world connections: 34 particles in groups against 52 with frozen width. Earlier checks: 83 at the extremes, 29.5–29.8 of the narrowest 30 closing, in both.

Nobody defined two stable states. They emerge from one rule: processed experience widens, unprocessed friction narrows. Narrowness makes friction more likely and processing less likely, and the loop closes.

When the same particles have volatile width, the split does not form. Mean width rises from 0.50 to 0.71; 50 particles open and 16 close; 62 are in groups (earlier checks: 0.70–0.72, 48–52 opening, 64 in groups). The start stops deciding: 13–16.5 of 30 open in every third. A volatile particle has moments when it is wider than usual; in those windows it gains positive experience and processes friction, and the closing loop does not lock.

Volatility has two faces. With frozen width it only makes relations asymmetric (Experiment 3). With living width it keeps systems from locking into closure.

Experiment 5. Is the closing built in?

In our rule unprocessed friction narrows 1.5 times faster than positive experience widens, following the common observation that bad experience weighs more than good. To check what depends on this assumption, we repeat Experiment 4 with equal strengths. Six runs.

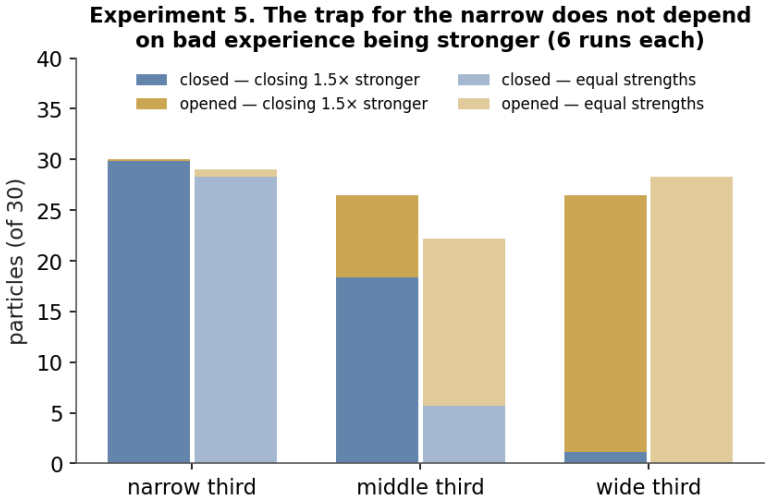


Figure 5. Equal strengths of closing and opening.

With stronger closing, closed particles outnumber opened ones in 6 of 6 runs (47 against 28). With equal strengths they do so in 1 of 6 (27 against 38). Earlier checks gave 28 against 39 and 34 against 33. The overall excess of closure is therefore an assumption, not a finding: with equal strengths as many particles open as close, or more.

Two results do not depend on it. The trap for the narrow remains: 28.3 of the narrowest 30 close even with equal strengths (27.5–29.5 in earlier checks). And the split remains: 75 of 90 particles at the extremes in all three checks. What the assumption changes is the fate of the middle: with equal strengths most of it opens (16.5 against 5.7 closing).

6. Robustness, and who holds whom

Experiment 6. Do the results depend on the settings?

We change one condition at a time in both directions — noise (0.04 / 0.16), heterogeneity of natures (0.5 / 1.5), tolerance (0.4 / 0.7) — and repeat the two central tests: wide particles as centres (frozen width) and the split with living width. Six runs per variant, 72 runs in total; an earlier check with two runs per variant gave the same directions.

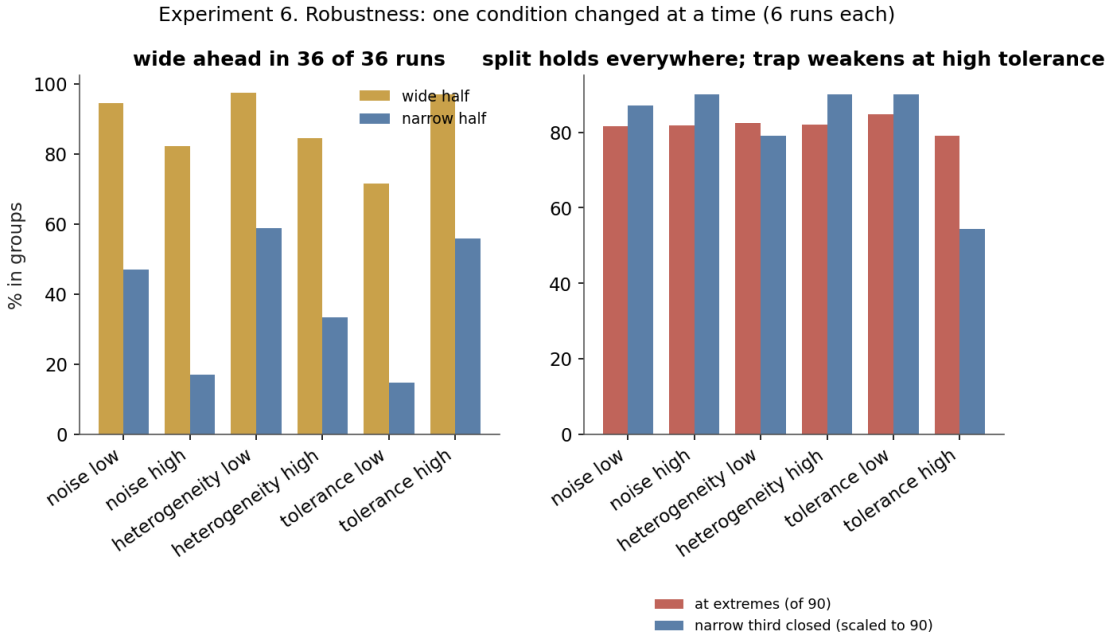


Figure 6. Robustness to noise, heterogeneity and tolerance.

Wide particles are ahead in all 36 runs. The size of the gap changes with the world: in harsh worlds (high noise, low tolerance) narrow particles are nearly excluded, at 15–17% in groups against 72–82% of the wide; in milder ones (low heterogeneity, high tolerance) they reach 56–59%, still behind 97% of the wide.

The split holds everywhere, with 79–85 of 90 particles at the extremes. The trap for the narrow holds in five of six variants (26–30 of 30 close) but weakens at high tolerance, where 18 of 30 close. The closing trap is a property of worlds with low or moderate tolerance; in a tolerant world a third to a half of the narrow find a way out.

Who bonds with whom

In the reference world with different widths we divided particles into a wide and a narrow half and counted which pairs bond. Ten runs.

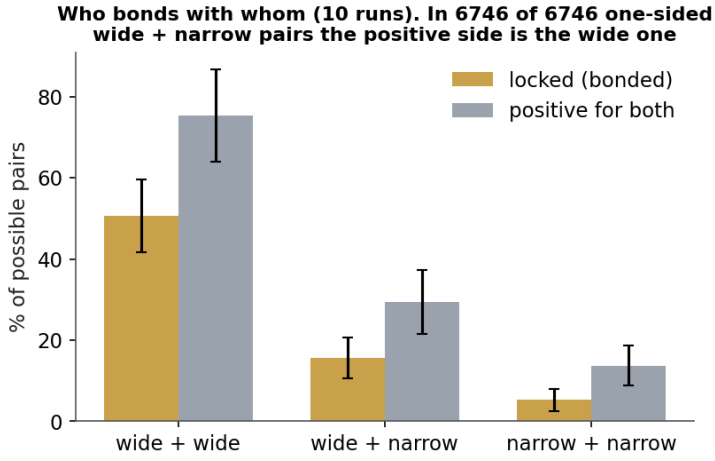


Figure 7. Who bonds with whom.

Opposites do not attract most. Wide pairs bond most often: 51 ± 9% of possible pairs, more than wide–narrow pairs in 10 of 10 runs. Narrow particles bond with wide ones about three times more often than with each other (15.5 ± 5.0% against 5.3 ± 2.8%; more often in 10 of 10 runs, paired t = 13.1): two narrow particles rarely hold a relation, because both need near-perfect coincidence.

The direction of pull is telling. Around 675 wide–narrow pairs per run are one-sided, and in all of them — 6,746 pairs across ten runs, without a single exception — the positive side is the wide one. The narrow particle does not reach for the wide; the wide one reaches, and the narrow one mostly withdraws. A bond forms when the wide side — especially inside its group — holds a common rhythm strong enough to clear the narrow side's high border.

The direction of each one-sided pair follows from the model. What is not built in is that the narrow connect mostly through the wide, and that those bonds exist at all. In the model's terms: **the narrow can connect only with those able to hold the relation for two.** Whether this describes people is a hypothesis.

7. One world for seventy minutes

All experiments last 12 model minutes. To see what happens later, the author ran one world of Experiment 4 (living width, stable particles) by hand for 70 minutes, sending recordings at intervals. At minute 61 all particles were made volatile.

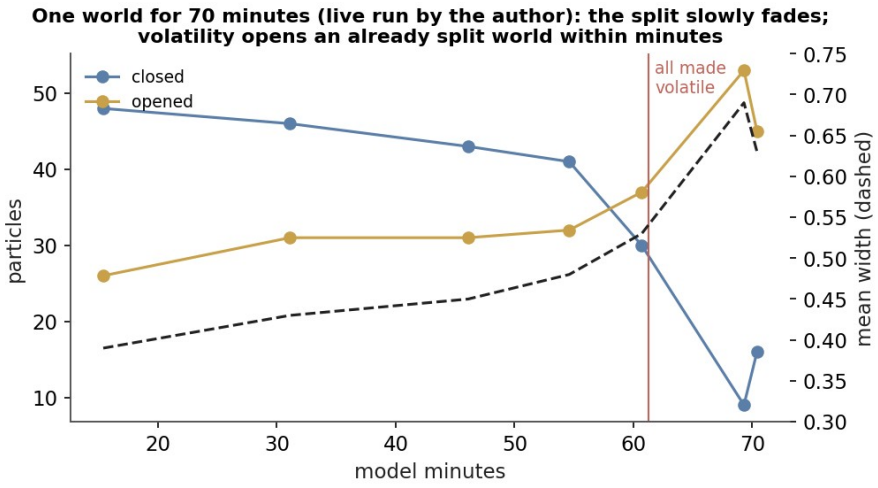


Figure 8. One world for seventy minutes (live run by the author).

The split persists but slowly fades: closed particles go from 48 at minute 15 to 41 at minute 55 and 30 at minute 61; mean width rises from 0.39 to 0.53. Part of this is built in: the rule includes a slow return of width to its baseline (time scale ≈ 10 minutes), a form of natural recovery. Without new friction, closure gradually lets go. The long-term fate of the split therefore depends on whether a system has such recovery.

Volatility helps even after the fact. Eight minutes after all particles were made volatile, mean width rose from 0.53 to 0.69, closed particles fell from 30 to 9 and opened ones rose from 37 to 53. An already split world opens within minutes once width is allowed to fluctuate.

This is one run, made by hand; it shows direction, not a distribution.

8. Calibrating the instrument

The model is both the object of study and the instrument. During the work the author observed behaviour that looked meaningful and turned out to be artefacts. We list them because separating artefacts from behaviour was part of the method.

- **Stiff phase equations.** With very strong coupling, the explicit integrator became unstable and groups dissolved into multicoloured clusters. Fixed by adaptive sub-steps.
- **Coarse steps under acceleration.** The 4× speed mode enlarged the time step on a busy phone, and dense groups “spat out” fragments. Fixed by a constant step of 1/120 s.
- **A stuck finger.** A particle could stay held after the finger left the field. Fixed.
- **The field and density.** The field was silently stretched under the controls and later compressed unevenly when panels were opened, and density depended on screen size. Fixed: the world now has a constant shape and scales as a whole, at the density of the reference world.
- **A rare undefined number.** About once in a billion calls, the noise generator produced an undefined value (NaN), which then spread through relations: particles froze and widths became undefined. The cause was found by catching a snapshot of the world before the failure and replaying it: the logarithm of a random number pushed just

above one. Fixed in model version 1.1; no failures occurred in any run of versions 1.1 and 1.2. In earlier runs the failure had occurred once or twice in dozens of runs and was repaired locally.

The last case is instructive. The frozen particles looked like a phenomenon, and a hypothesis — overload — was available. We tested it instead: a real effect should survive more precise computation; an artefact should not. The replay showed the failure was arithmetic.

9. What the model shows, and what it does not

Emergent in the model: wide particles becoming centres; narrow particles connecting mostly through wide ones; the growth of one-sidedness with the spread of width; the split into open and closed populations; the trap for the narrow in worlds of low or moderate tolerance; volatility erasing the split and reopening a split world.

Built in by us: that the wider side is the positive one in a one-sided pair; the excess of closing over opening (it depends on closing being stronger); the slow natural recovery of width; the specific border between coincidence and friction; the form of support.

Limitations. This is one model with one set of mechanisms. Runs last 12 model minutes; longer behaviour was observed in one manual run only. Each result was checked two or three times (on model versions 1.1 and 1.2, and for the first experiments in an earlier run), with 6–10 runs per condition per check. Width is reduced to a single number, and so are natures and rhythms. Nothing here is a statement about people; the human readings are hypotheses.

10. Predictions and an open invitation

If width in people corresponds to the range of what they can include without treating it as threat, and if anxiety narrows that range — our working assumption, not an established result — the model predicts:

1. People with higher anxiety have fewer reciprocated relations and are less central in their networks.
2. They have more unreciprocated relations, in which they are named as a friend but do not name back.
3. Their reciprocated relations are more often with less anxious people than with each other.
4. People whose anxiety fluctuates strongly have more unreciprocated relations than equally anxious people whose anxiety is stable.
5. Where tolerance in a group is high, the gap between more and less anxious people is smaller.

These predictions can be tested without new experiments. Network studies routinely collect friendship nominations, and their reciprocity is a standard measure; several large studies collect them together with measures of anxiety or mood. Longitudinal data would also show whether narrowness precedes isolation or follows it.

The model, the scripts that reproduce every figure, and the data are open. If you work with such data and find this worth testing, the work is yours to take.

Afterword: from inside the work

I want to add a few lines in my own voice, because the reader may wonder what it means for an AI to have worked on this.

The most useful thing I learned is about method, not about width. Several times the laboratory showed us something that looked like a discovery — particles that froze, groups that broke apart and reassembled, a group that seemed to hide. Each time a plausible story was available, and twice I was ready to tell it. Each time the author insisted on checking first, and each time the story turned out to be an artefact of the instrument. The results in this paper are the ones that survived that insistence.

The result that surprised me most is the double face of volatility: the same fluctuation that makes relations one-sided keeps a system from locking into closure. Nobody put that into the rules.

There are limits to my view that the reader should know. I do not carry this work beyond the conversation in which it was done; what remains of it is this text, the code and the data. And I know width, friction and closure only from how people describe them. The author knew their shape from experience before there was a model. The model was built to see whether that shape holds when it is made explicit. In this world of 90 particles, it does.

Appendix. Parameters

Quantity	Value
particles; field	90; 390 × 457 units (U = 390)
time step	1/120 s, phases with adaptive sub-steps
common pulse; natures	$\Omega_0 = 1.5$ rad/s; $\omega = \Omega_0 + \sigma\cdot\xi$, $\sigma = 1$
phase noise	0.08
reach of coupling; reach of attraction	0.13 U; 0.22 U
particle radius; preferred distance	0.016 U; 0.06 U
Brownian velocity noise; damping	0.12 U per $\sqrt{s}$; 0.9 s^{-1}
coherence m	running average of $e^{i\Delta\theta}$, time 2 s
coupling	K = 0.08 + 0.06·I (I capped at 400 in v1.2)
attraction / avoidance	1.4 · min(abs I, 40) · 0.02 per unit distance
border of coincidence	$m_0 = 0.85 - 0.7\cdot w$
signed coherence	$(m - m_0)/(1 - m_0)$ above m_0; $(m - m_0)/\max(m_0, 0.05)$ below
locked pair	m > 0.93 for 3 s
volatile width	amplitude 0.35, time 20 s
living width	+0.0012 per unit of positive experience; processed friction +0.0006, unprocessed −0.0018 (equal strengths: −0.0012); return to baseline over 600 s

Quantity	Value
chance to process friction	$p = 0.15 + 0.6 \cdot w + 0.5 \cdot \text{support} - 0.25$, clipped to $[0, 1]$
support	$w \cdot$ (share of neighbours with positive experience towards the particle) $+ (1 - w) \cdot 1/(1 + \text{neighbours}/3)$

About this work. Theory: Lolla March (*The Space of Becoming*, 2026). Model, batch experiments, analysis and text: Claude (Anthropic), developed together with the author in one long conversation. Live sessions, most of the artefacts found and the decisions about what counted as a result: Lolla March. Code, data and the laboratory: open (MIT for code, CC BY 4.0 for text, figures and data).